TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Gunnar Walther	Email	gunnar.walther@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer Institute for Manufacturing Technology and Advanced Materials IFAM, Dresden Branch			
Address	Winterbergstr. 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	Iron as stationary and mobile storage of hydrogen for the future circular renewable energy economy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), **Hydrogen (√),** Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Energy & Resources		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Metal Materials	Energy Materials Technologies
Concept of Proposed Technology	The proposed technology is based on the use of **iron as a solid-state energy carrier for hydrogen**, exploiting the reversible iron–iron oxide redox cycle. Iron offers a unique combination of properties that make it particularly suitable for large-scale energy storage and transport in a future renewable energy system. First, **iron is a low-cost and globally abundant raw material**. While hydrogen itself is not a primary energy source but an energy carrier, iron enables hydrogen to be stored in a chemically bound, solid form with a **very high volumetric energy density of approximately 9,700 Wh/L**. This value is significantly higher than that of compressed hydrogen (≈2,100 Wh/L at 700 bar) and orders of magnitude higher than gaseous hydrogen under ambient conditions (≈3 Wh/L). As a result, iron-based storage enables compact, safe, and space-efficient energy storage and transport. Second, iron-based hydrogen storage relies on **well-known, robust, and industrially established**			

	processes, such as hydrogen reduction and water–gas reactions, which have been applied in metallurgy and chemical engineering for decades. This makes the technology inherently **cost-effective, scalable, and manageable from an industrial safety perspective**, especially when compared to cryogenic or ultra-high-pressure hydrogen systems. Third, the proposed concept strongly supports **circular economy principles**. Large quantities of iron oxide are generated as industrial by-products, particularly in the steel industry during pickling processes, where iron oxide-rich pickling sludge is produced. This material is already available in fine, powder-like form and can be directly utilized as feedstock for hydrogen storage materials. The recycling of such secondary raw materials reduces waste, conserves primary resources, and significantly lowers the environmental footprint of the technology. Overall, iron combines **high energy density, low cost, global availability, recyclability, and technological maturity**, making it an ideal candidate for large-scale hydrogen storage and transport within a renewable energy-based economy.
Rationale for Joint R&D Project	The need for this joint R&D project arises from the urgent requirement to overcome the **storage and transport limitations of hydrogen**, which currently hinder its widespread adoption in renewable energy systems and industrial decarbonization. Although electrolysis technologies are rapidly advancing, the downstream handling of hydrogen remains costly, energy-intensive, and infrastructure-heavy. Both Germany and Korea are highly industrialized countries with strong commitments to carbon neutrality and hydrogen-based energy strategies. Importantly, **Korea is also a major steel producer**, with extensive steel processing infrastructure and a significant availability of iron oxide-containing by-products such as pickling sludge. Furthermore, Korea has a strong industrial base in **raw material processing, reactor engineering, and system manufacturing**, making it an ideal partner for the development and industrial implementation of iron-based hydrogen storage technologies. The collaboration is necessary because the project spans the full innovation chain, from **materials science and powder metallurgy** to **reactor design, system integration, and industrial validation**. The German partner contributes expertise in iron oxide processing, redox cycling analysis, and material stabilization, while the Korean SME brings application-driven development, engineering capabilities, and insights into industrial deployment and commercialization. This complementary expertise ensures that the project results are not only scientifically sound but also industrially relevant and scalable.
Key Words	1. Iron-based solid-state hydrogen storage 2. Iron–iron oxide redox energy carrier 3. High-density hydrogen transport medium 4. Circular renewable energy systems 5. Industrial hydrogen logistics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5
R&D Objectives	This R&D project aims to develop **cycle-stable iron-based hydrogen storage materials and associated process technologies** based on the reversible iron–iron oxide redox cycle. At the **start of the project (TRL 3)**, the underlying redox reactions between iron, iron oxides, hydrogen, and water vapor are scientifically well understood and experimentally demonstrated at laboratory scale. However, the technology has not yet been validated for practical hydrogen storage applications, particularly with respect to long-term cycling stability, material morphology control, and reactor-relevant operating conditions. The project is designed to advance the technology to **TRL 5 at project completion**, where the developed materials and processes will be validated in a relevant environment. This includes the development of tailored iron oxide powders, granules, and pellets with defined porosity and mechanical stability, the identification of optimal reduction and oxidation process windows, and the demonstration of stable redox cycling under application-relevant temperatures, gas compositions, and reactor concepts. Specific technologies to be developed include: • Processing of iron oxide powders derived from industrial secondary raw materials, such as pickling sludge from steel processing • Granulation and pelletization techniques to create mechanically stable particles with defined porosity and gas permeability in the fluidized bed • Stabilization concepts using inert oxides, alloying elements, and spinel-forming phases to suppress sintering and degradation • Optimized hydrogen reduction and water vapor oxidation processes, including control of hydrogen and steam partial pressures • Oxidation of iron using water vapor for efficient hydrogen release • Evaluation of reactor concepts suitable for cyclic redox operation These technologies aim to combine high reactivity with long-term mechanical and chemical stability. By the end of the project, the technology will have progressed from a laboratory proof of concept to a **validated hydrogen storage solution under relevant operating conditions**, providing a robust foundation for subsequent pilot-scale demonstrations targeting TRL 6.			
R&D Target Specifications	The project targets the following quantitative performance indicators: • Volumetric energy density: **up to 9,700 Wh/L (iron-based storage material)** • Effective hydrogen storage capacity: **≥3 wt% H_2 (system level)** • Redox cycling stability: **>500 cycles with less than 10% performance loss** • Operating temperature window: **600–800 °C** • Hydrogen purity during release: **>99.9%** • Utilization of recycled iron oxide feedstock: **>70%** • Mechanical integrity of pellets/granules after cycling: **>90% mass retention**			

R&D Plan	The research program is structured into two consecutive phases with multiple work packages (WPs) with clearly defined objectives, methodologies, and milestones. **Phase 1 (Months 1–6)** focuses on the **feasibility assessment** of the proposed iron-based hydrogen storage concept. Based on the results of Phase 1, a formal decision will be taken on whether the project proceeds to **Phase 2 (Months 7–36)**, which aims at technology development and validation up to TRL 5. **<u>Phase 1: Feasibility Demonstration (Months 1–6)</u>** **Objectives:** To demonstrate the fundamental technical feasibility of using iron oxide-based materials derived from primary and secondary feedstocks for reversible hydrogen storage via the iron–iron oxide redox cycle. **Assigned Work Packages:** **WP1: Raw material selection and characterization (Months 1–6)** Chemical, physical, and structural characterization of iron oxide feedstocks, including pickling sludge-derived materials. **WP2 (initial): Material processing and morphology screening (Months 3–6)** Preliminary granulation and shaping trials to assess processability and basic porosity formation. **WP3 (initial): Redox behavior screening (Months 4–6)** First reduction/oxidation tests using DTA–TG–MS to verify reversible hydrogen storage behavior and reaction kinetics. **Milestone after Phase 1** **M1: Feasibility milestone and decision on Phase 2 funding** **Month 6** Successful demonstration of reversible iron–iron oxide redox behavior with measurable hydrogen uptake and release, definition of at least one technically viable material concept, and confirmation of achievable operating temperature ranges. **<u>Phase 2: Technology Development and Validation (Months 7–36)</u>** **Objectives:** To develop, optimize, and validate cycle-stable iron-based hydrogen storage materials and processes under application-relevant conditions, advancing the technology from TRL 3 to TRL 5. **Assigned Work Packages:** **WP2 (continued): Material processing and morphology control (Months 7–12)** Optimization of granulation, pelletization, and porosity control strategies. **WP3 (continued): Redox behavior and thermochemical analysis (Months 7–18)** Detailed kinetic and thermodynamic studies to define optimal reduction and oxidation windows. **WP4: Stabilization strategies and cycling stability (Month 12–27)** Materials will be modified using inert oxides, alloying elements, and spinel-forming additives to suppress sintering and structural degradation. Long-term redox cycling experiments will be conducted to assess durability.

WP5: Reactor-relevant testing and process integration (Month 18–33)

Selected material formulations will be tested under conditions representative of fixed-bed or moving-bed reactors. Heat and mass transfer behavior will be evaluated.

WP6: Evaluation and optimization (Month 30–36)

Performance data will be analyzed to define optimal material compositions, process windows, and design guidelines for scale-up.

WP7: Exploitation, Valorization, and Market Preparation (Month 12-36)

Objectives:

The objective of WP7 is to ensure the **systematic exploitation and valorization of the project results**, bridging the gap between laboratory-scale research and industrial application. WP7 focuses on intellectual property (IP) management, market-oriented evaluation of the developed technologies, and preparation for commercialization in relevant industrial sectors.

Scope and Activities:

WP7 will analyze potential exploitation pathways for the developed iron-based hydrogen storage materials, processes, and reactor concepts. This includes the identification of **application-specific use cases**, such as hydrogen supply for steelmaking, chemical production, and decentralized energy storage. Techno-economic assessments will be conducted to benchmark the proposed technology against conventional hydrogen storage and transport solutions.

A structured **IP and know-how management strategy** will be implemented. Background and foreground IP will be clearly defined, and opportunities for patents, utility models, or trade secrets will be evaluated jointly by the project partners. Particular emphasis will be placed on material formulations, stabilization concepts, and process parameters developed within the project.

Market and Business Development:

Market readiness will be assessed through performance, cost, and scalability analyses. Potential business models include:

- commercialization of iron oxide-based storage materials (powders, granules, pellets),
- licensing of reduction/oxidation process know-how and reactor concepts,
- joint development of modular hydrogen release systems for industrial users.

The Korean SME will contribute market insights, customer requirements, and industrial validation scenarios, while the German research partner will support technology assessment and documentation.

This will result in a **clear exploitation roadmap**, reducing time-to-market and enabling follow-up pilot and demonstration projects. It will ensure that the project outcomes have a high probability of industrial uptake and long-term economic impact.

Additional Milestone within Phase 2

M2: Demonstration of cycle-stable material performance under relevant conditions
Month 24

Demonstration of significantly enhanced cycling stability (>100–300 redox cycles) for optimized material formulations, confirming suitability for reactor-scale validation.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The German research institute (IFAM) will focus on **materials development, powder processing, and fundamental redox characterization**. Tasks include: • Processing of iron oxide powders from industrial by-products • Development of morphology-controlled iron and iron oxide materials • Granulation and pelletization with controlled porosity • Thermoanalytical redox studies (DTA–TG–MS) • Microstructural and metallographic analysis • Development of stabilization strategies via alloying and spinel formation IFAM provides specialized laboratory infrastructure and extensive experience in powder metallurgy and hydrogen-based reduction processes.
Expected R&R of Korean SME Partner	The Korean SME will focus on applied development, reactor engineering, and validation. Responsibilities include: • Design and evaluation of suitable reactor concepts • Testing of developed materials under realistic operating conditions • Assessment of heat and mass transfer behavior • Evaluation of scalability and manufacturability • Contribution to system-level performance analysis • Preparation for commercialization and industrial deployment The SME will provide engineering resources, pilot-scale testing capabilities, and industrial development expertise.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver significant technological, industrial, and societal outcomes by advancing iron-based hydrogen storage from **laboratory proof-of-concept (TRL 3)** to a **technology validated under relevant operating conditions (TRL 5)**. In doing so, the project addresses one of the most critical bottlenecks of the hydrogen economy: the **extremely low volumetric energy density of hydrogen**, which is approximately 3 Wh/L under ambient conditions and requires energy-intensive compression or liquefaction to achieve practical storage densities. By chemically storing hydrogen in iron, volumetric energy densities of up to **approximately 9,700 Wh/L** can be realized, enabling compact, safe, and cost-efficient energy storage and transport. At **TRL 5**, the project will deliver **cycle-stable iron oxide-based storage materials** with tailored morphology, controlled porosity, and enhanced resistance to sintering and structural degradation during repeated redox cycling. Stabilization strategies based on inert oxides, alloying elements, and spinel-forming phases will significantly improve material durability and operational robustness. In parallel, validated reduction and oxidation process windows will provide a sound basis for reactor design and system integration under industrially relevant conditions. These outcomes form a critical enabler for **industrial decarbonization**, particularly in sectors with high hydrogen demand. In the steel industry, reliable hydrogen availability is a prerequisite for the transition from coke-based blast furnace routes to **hydrogen-based direct reduction of iron (DRI)**. By reaching TRL 5, the project demonstrates that iron-based hydrogen storage and transport without reliance on high-pressure tanks or cryogenic infrastructure is technically viable under conditions relevant to industrial operation, thereby reducing technological and economic barriers for adoption and supports substantial CO_2 emission reductions. Similar benefits apply to the chemical industry, where fossil-based hydrogen can be replaced by green hydrogen supplied via solid iron energy carriers. The commercialization potential of the project is explicitly addressed through **WP7 (Exploitation, Valorization, and Market Preparation)**. WP7 ensures that the developed materials, processes, and system concepts are systematically evaluated with respect to techno-economic feasibility, scalability, and market relevance. Exploitation pathways include the commercialization of iron oxide powders, granules, and pellets as hydrogen storage media, as well as the licensing of process know-how and reactor concepts. The use of recycled iron oxide feedstocks, such as pickling sludge, provides a strong cost advantage and reinforces circular economy principles. Beyond the project duration, the validated **TRL 5 technology** establishes a clear and credible pathway toward **pilot-scale demonstration and system integration at TRL 6**, which can be addressed in targeted follow-up projects. This transition is explicitly supported by **WP7**, which establishes a clear commercialization and exploitation roadmap, including intellectual property management, identification of early adopters, and preparation of pilot and demonstration activities. Through the close involvement of the Korean SME as an industrial development and system integration partner, the project outcomes are expected to achieve accelerated market entry and industrial uptake. In the long term, the project enables the development of **iron-based energy carrier supply chains**, in which reduced iron is produced at locations with abundant renewable energy and transported safely and efficiently to industrial users for on-demand hydrogen release. Supported by the exploitation activities defined in WP7, this concept opens new business models in hydrogen logistics, strengthens international energy trade opportunities, and positions **Germany and Korea as key contributors and technology leaders** in the global transition toward sustainable hydrogen and energy storage technologies.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization